# Supplement.

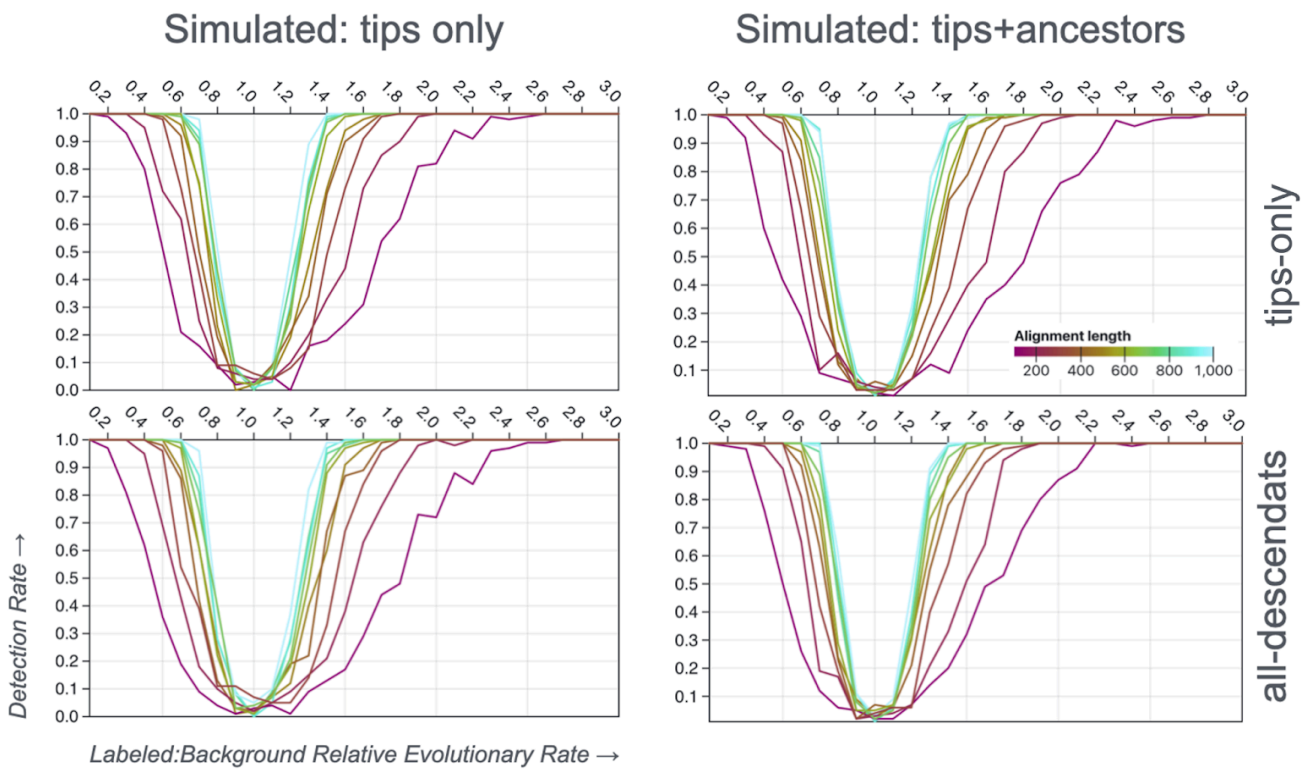


**Figure S1.** Detection rate (fraction of 100 replicates with p-value ≤0.05 for a given scenario) for the `Proportional:Unconstrained test` test, as a function of simulated alignment length (color), the labeled/background branch rate ratio (**$R_L$ varied,** $R_0$ = 1, x-axis), and the labeling strategy (`all-descendants`, or `tips-only`; table row). We considered simulations where the labeled set was just the terminal branches (tips only, Fig 1B) or terminal branches and their ancestors (Fig 1C). The `WAG` evolutionary model was used to simulate and infer.

Alt text: Four small plots showing detection rate for the `Proportional:Unconstrained test` test on the y-axis versus the labeled branch rate on the x-axis. The panels differ by simulation strategy (tips only or tips plus ancestors) and by labeling strategy (all-descendants or tips-only). Curves are V-shaped, with longer alignments producing narrower, more accurate curves, and mismatched labeling strategies producing slightly wider, less accurate curves, although not much difference is observed.

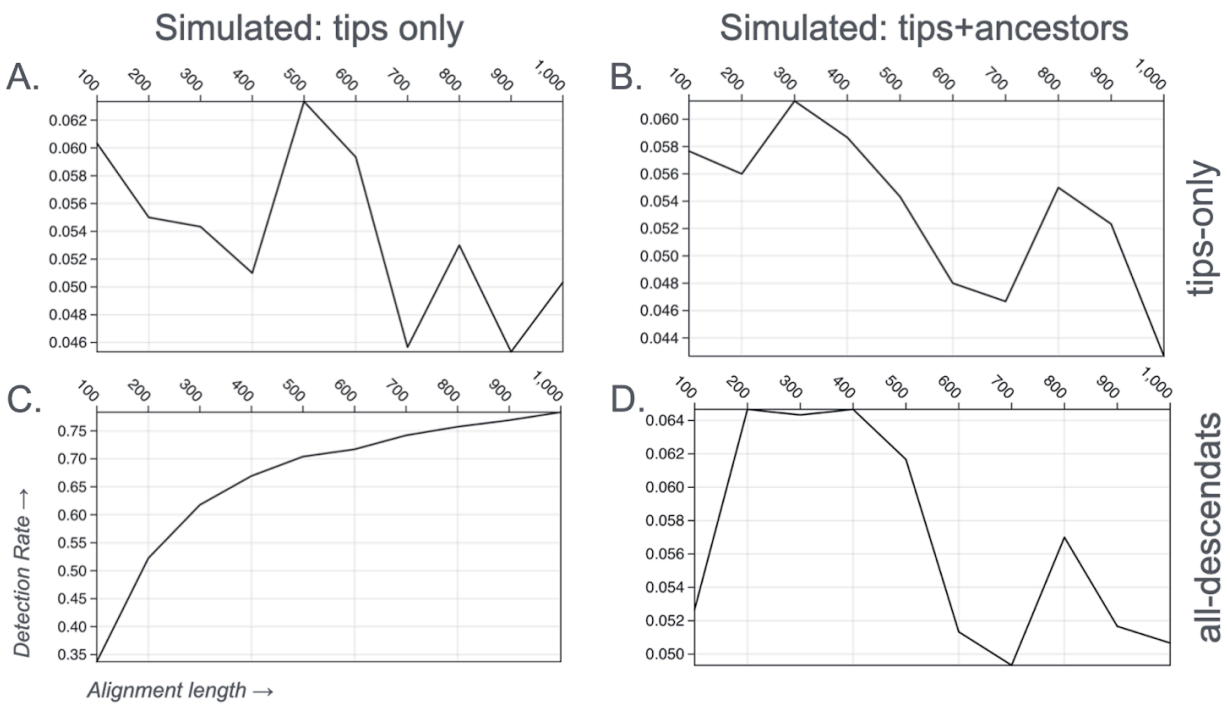


**Figure S2.** Detection rate (fraction of replicates with p-value ≤0.05 for a given set of scenarios) for the `Proportional Partitioned:Unconstrained test` test, which evaluates whether the branches in the labeled set evolve at rates that differ proportionally from the tree. Results are shown as a function of simulated alignment length (x-axis, aggregated over all non-null simulations, i.e. $R_L \neq 1$), and for two labeling strategies (rows): `tips-only` (no descendant branches labeled) and `all-descendants` (all descendant branches of labeled nodes included in the labeled set). We considered simulations where the labeled set was just the terminal branches ("tips only", Fig 1B) or terminal branches and their ancestors ("tips+ancestors", Fig 1C). All simulations and inferences used the `WAG` evolutionary model. Note the difference in the y-axis scale across the four plots, which highlights significant variation in detection rates depending on the simulation and labeling strategies.

Alt text: Four small plots showing detection rate for the `Proportional:Unconstrained test` test on the y-axis versus the labeled branch rate on the x-axis. The panels differ by simulation strategy (tips only or tips plus ancestors) and by labeling strategy (all-descendants or tips-only). In the tips-only simulation with all-descendants labeling, detection rate increases with alignment length and reaches about 0.80. All other plots all stay below 0.65 and are generally decreasing.

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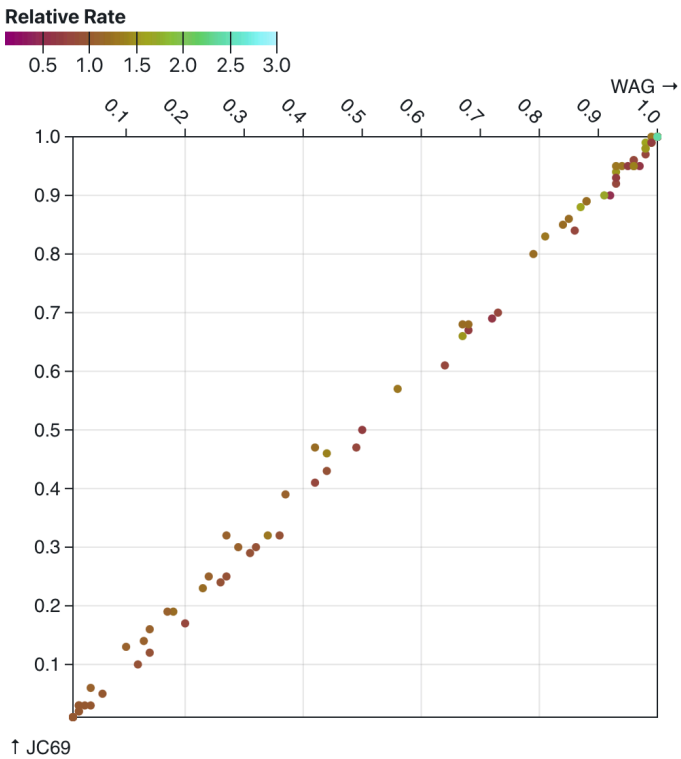



**Figure S3.** Detection rate (fraction of 100 replicates with p-value ≤0.05 for a given scenario) for the `Proportional:Proportional Partitioned` test, as of the labeled/background branch rate ratio (**$R_L$ varied, $R_0$** = 1, color), simulated and inferred using tips+ancestors (`all-descendants`, or `none`; table row). The plot compares paired detection rates (by simulation scenario) using the correct `WAG` substitution model or the incorrect `JC69` substitution model.

Alt text: Scatterplot comparing detection rates from `JC69` (x-axis) and `WAG` (y-axis) under the `Proportional:Proportional Partitioned` test. Points fall tightly along a 45° diagonal, indicating near-identical estimates between models. Point colors reflect relative rates from low (purple) to high (cyan), with higher relative-rate points clustering toward the upper end of the line.

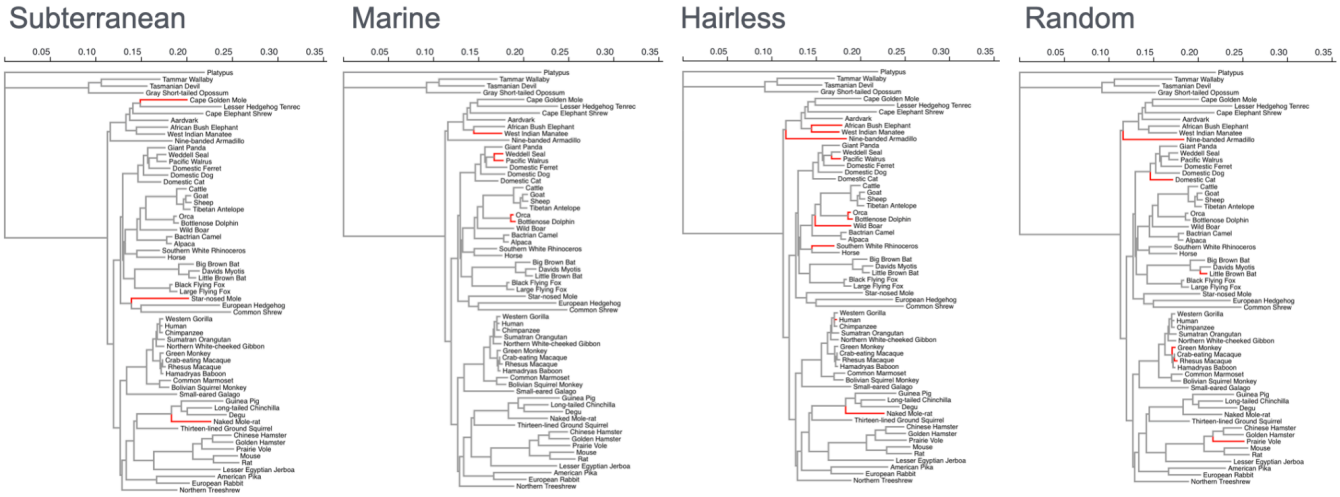
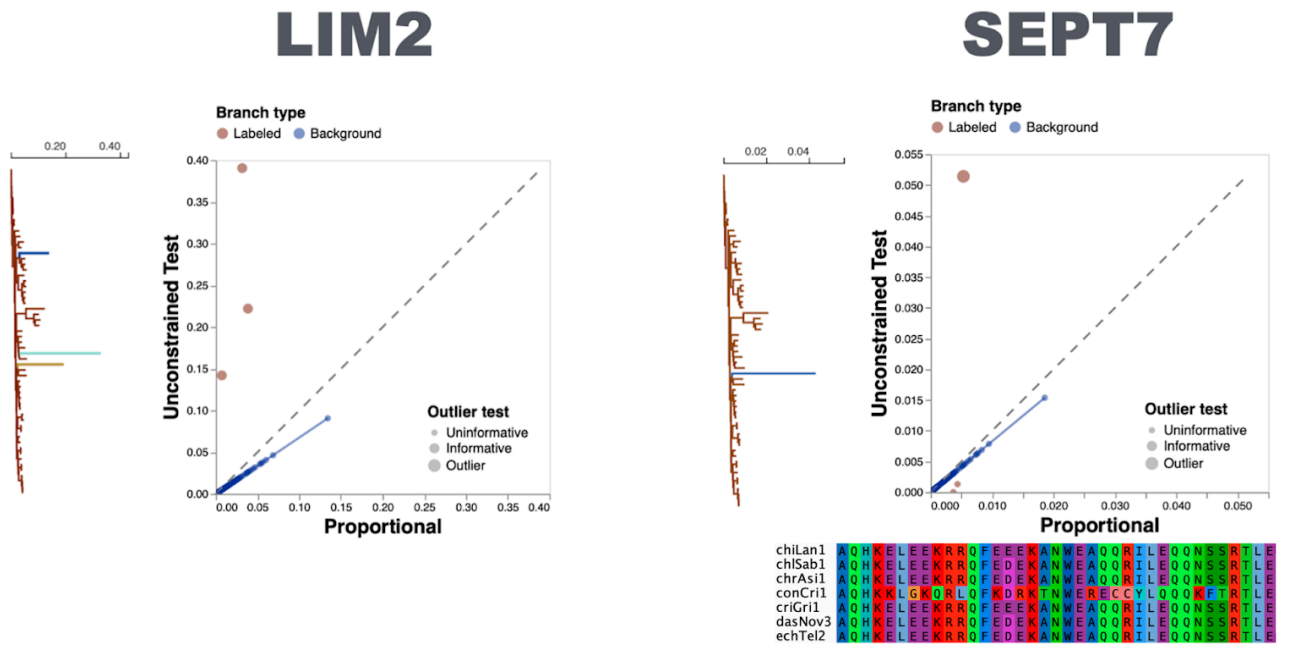



**Figure S4.** Reference tree for the UCSC mammalian species, with the four labeled subsets highlighted.

Alt text: Four phylogenies showing the labelings of the different phenotypes tested including subterranean mammals, marine mammals, hairless mammals, and random labeling. Labeled branches are shown in red.




| Gene | Branch | Only branch | All but this branch |
|---|---|---|---|
| LIM2 | Naked mole-rat | $<10^{-16}$ | $<10^{-16}$ |
| | Star-nosed mole | $10^{-14}$ | $<10^{-16}$ |
| | Cape golden mole | $<10^{-16}$ | $<10^{-16}$ |
| SEPT7 | Naked mole-rat | 0.28 | $\sim10^{-11}$ |
| | Star-nosed mole* | $\sim10^{-12}$ | 0.49 |
| | Cape golden mole | 0.32 | $\sim10^{-11}$ |

**Figure S5.** Two genes with a significant Proportional:Proportional Partitioned test result for the subterranean phenotype in UCSC data. *LIM2* derives its signal from all three tested branches, while *SEPT7* – only from the star-nosed mole branch. A manual inspection of the corresponding gene alignment shows a clear sequence assembly/alignment error in this species (conCri1) towards the 3' end.

Alt text: Two gene-specific panels, one for LIM2 (left) and one for SEPT7 (right), each showing (1) a tree with labeled (foreground) and background branch lengths, (2) a scatterplot comparing Proportional vs. Unconstrained branch-length estimates where labeled branches appear as brown points and background branches as blue points, with outliers marked by darker colors. SEPT72 also includes an amino acid alignment snippet, highlighting an assembly/alignment error in the species conCri1. Below the plots is a table listing, for each gene and each labeled subterranean species (naked mole-rat, star-nosed mole, Cape golden mole), the outlier test p-values for the "only branch" and "all but this branch" tests, showing very small p-values for most branches and indicating which branches are driving the outlier signal.

| Gene ID | Process | Branch Count | Relative Rate | RELAX K | RELAX p-value |
|---|---|---|---|---|---|
| *LIM2* | Lens fiber cell differentiation, cell adhesion | 3 | 14.78 | 0.257 | **0.00000000** |
| *CBLN4* | Synaptic organization, cerebellar development | 2 | 8.25 | 0.373 | **0.00000432** |
| *CRYBB3* | Lens structural protein, cataract formation | 2 | 7.42 | 0.273 | **0.00000000** |
| *CRYBB2* | Lens structural protein, cataract formation | 2 | 6.95 | 0.348 | **0.00000000** |
| *MIP* | Lens fiber cell transmembrane water transport, cell adhesion | 2 | 5.13 | 0.507 | **0.00000000** |
| *OR5I1* | Olfactory signaling | 2 | 4.91 | 1.002 | 0.98112737 |
| *OR6N2* | Olfactory signaling | 3 | 4.79 | 0.400 | 0.00052188 |
| *CRYGC* | Lens structural protein, cataract formation | 3 | 4.68 | 0.576 | **0.00000276** |
| *TUBB6* | Microtubule component, cytoskeleton | 3 | 4.52 | 0.524 | **0.00000000** |
| *MTCP1* | T-cell development, apoptosis (potentially) | 2 | 4.48 | 0.888 | 0.49845762 |
| *CRYGB* | Lens structural protein, cataract formation | 2 | 4.43 | 0.689 | 0.01514514 |
| *NQO2* | Quinone detoxification | 2 | 4.36 | 0.470 | 0.00004209 |
| *MNX1* | Motor neuron development, pancreas development | 2 | 3.99 | 0.994 | 0.98096523 |
| *OR8K1* | Olfactory signaling | 2 | 3.81 | 0.458 | 0.00022084 |
| *AK128361* | Unknown, predicted protein. | 3 | 3.80 | 1.088 | 0.74349370 |
| *ARHGDIA* | Rho GTPase regulation, vesicle trafficking | 2 | 3.79 | 0.808 | 0.00552766 |
| *GUCA1B* | Phototransduction, calcium regulation in photoreceptors | 3 | 3.72 | 0.296 | **0.00000000** |
| *OR5AR1* | Olfactory signaling | 2 | 3.68 | 1.199 | 0.20261633 |
| *OR6K2* | Olfactory signaling | 2 | 3.66 | 0.722 | 0.00156589 |
| *OR6C6* | Olfactory signaling | 2 | 3.57 | 0.721 | 0.00419302 |
| *TACR3* | Neuropeptide signaling, smooth muscle contraction | 2 | 3.48 | 1.079 | 0.30014599 |
| *EI24* | Autophagy regulation, response to cellular stress | 2 | 3.45 | 0.448 | **0.00000002** |
| *CABP5* | Calcium signaling in photoreceptors | 2 | 3.42 | 0.000 | **0.00000000** |
| *TAAR8* | Trace amine-associated receptor, olfactory signaling | 2 | 3.33 | 1.194 | 0.08286307 |
| *OR2T2* | Olfactory signaling | 3 | 3.32 | 0.376 | **0.00000000** |
| *PRTFDC1* | Peptide hormone processing, secretory pathway | 3 | 3.32 | 0.189 | **0.00000000** |
| *LGSN* | Proteolysis, antigen presentation | 3 | 3.32 | 0.166 | **0.00000000** |
| *OR13C2* | Olfactory signaling | 3 | 3.26 | 0.945 | 0.53033301 |
| *GJC1* | Gap junction communication | 2 | 3.12 | 0.687 | 0.00007454 |
| *ZNF404* | Transcription regulation (likely) | 2 | 3.12 | 0.842 | 0.37627799 |
| *HSF4* | Transcription factor, lens development, heat shock response | 2 | 3.08 | 0.486 | **0.00000000** |
| *PCDHB19P* | Cell adhesion (Potentially a Pseudogene) | 2 | 3.08 | 0.886 | 0.37074667 |
| *HIGD1C* | Mitochondrial function, response to hypoxia | 2 | 3.08 | 0.070 | 0.00008967 |
| *CTRB2* | Serine protease, digestion | 2 | 2.86 | 0.679 | 0.03687959 |
| *AMN1* | Kidney development, vitamin B12 absorption | 2 | 2.84 | 0.607 | 0.00002108 |
| *ROM1* | Photoreceptor outer segment structure | 3 | 2.84 | 0.489 | **0.00000000** |
| *FAM172BP* | Unknown. Predicted protein. Pseudogene. | 2 | 2.82 | 0.000 | 0.00004434 |

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|---|---|---|---|---|---|
| *OR4X2* | Olfactory signaling | 2 | 2.77 | 0.595 | **0.00000099** |
| *NCR2* | Natural killer cell cytotoxicity | 2 | 2.73 | 0.676 | 0.25247365 |
| *PCDHB18* | Cell adhesion | 2 | 2.72 | 0.423 | **0.00000003** |
| *OPN1MW* | Color vision (medium wavelength) | 2 | 2.72 | 0.671 | 0.00004080 |
| *TEX12* | Germ cell development, meiosis | 2 | 2.69 | 0.036 | 0.00206887 |
| *TSPAN1* | Cell adhesion, cell motility, membrane organization | 2 | 2.69 | 0.000 | **0.00000002** |
| *MITD1* | Endosomal trafficking | 2 | 2.69 | 0.600 | 0.00080794 |
| *DSCAM* | Neuronal development, cell adhesion | 2 | 2.68 | 0.769 | **0.00000000** |
| *SERINC4* | Serine incorporation into membranes, lipid metabolism | 3 | 2.64 | 0.310 | **0.00000000** |
| *RAX2* | Eye development, photoreceptor differentiation | 2 | 2.63 | 0.518 | 0.00001293 |
| *LENEP* | Regulation of ER stress response | 2 | 2.62 | 0.276 | 0.00253362 |
| *PBX4* | Transcription factor, spermatogenesis (likely) | 2 | 2.61 | 0.470 | 0.00010895 |
| *KRT81* | Hair keratin, hair follicle differentiation | 3 | 2.61 | 0.707 | **0.00000012** |
| *PAPD7* | RNA processing, non-canonical poly(A) polymerase | 2 | 2.59 | 0.865 | 0.03901335 |
| *FOXE3* | Lens development, transcription factor | 2 | 2.57 | 0.631 | 0.00051118 |
| *EXT1* | Heparan sulfate biosynthesis | 2 | 2.52 | 0.667 | 0.01597876 |
| *FADS2* | Fatty acid metabolism, desaturation | 3 | 2.51 | 0.934 | 0.41365111 |
| *ZNF614* | Transcription regulation (likely) | 3 | 2.50 | 0.782 | 0.04698792 |
| *NXPE4* | Neuropeptide processing (potentially) | 3 | 2.48 | 0.710 | 0.12188795 |
| *PRSS56* | Serine protease, retinal development (likely) | 2 | 2.47 | 0.462 | **0.00000000** |
| *LITAF* | Regulation of TNF production, lysosomal degradation | 2 | 2.46 | 0.534 | 0.00293967 |
| *SGMS2* | Sphingolipid metabolism | 3 | 2.45 | 0.774 | 0.01389908 |
| *FADS3* | Fatty acid metabolism, desaturation (likely) | 2 | 2.42 | 0.922 | 0.40120868 |
| *NRL* | Photoreceptor development, transcription factor | 2 | 2.42 | 0.653 | 0.00022682 |
| *PDE6B* | Phototransduction, cGMP hydrolysis | 2 | 2.38 | 0.672 | **0.00000017** |
| *KRT33B* | Hair keratin, hair structure | 2 | 2.38 | 0.646 | **0.00000715** |
| *ZNF568* | Transcription regulation (likely) | 2 | 2.35 | 0.680 | 0.00406186 |
| *KRT12* | Corneal epithelial keratin | 2 | 2.34 | 0.396 | **0.00000001** |
| *GPR34* | Immune response, macrophage function | 2 | 2.33 | 0.465 | **0.00000004** |
| *SETD7* | Histone methylation, transcriptional regulation | 2 | 2.32 | 0.724 | 0.00037011 |
| *ERG* | Transcription factor, endothelial cell development, hematopoiesis | 2 | 2.30 | 0.791 | 0.00660334 |
| *KRT20* | Intestinal epithelial keratin | 2 | 2.30 | 0.731 | 0.12303507 |
| *OR52E8* | Olfactory signaling | 2 | 2.29 | 0.981 | 0.83108921 |
| *OR10J5* | Olfactory signaling | 3 | 2.29 | 0.302 | **0.00000000** |
| *FAM200B* | Bone and tooth development | 2 | 2.28 | 0.455 | 0.00005190 |
| *ITLN2* | Innate immunity, inflammatory bowel disease (potential) | 3 | 2.27 | 0.000 | **0.00000290** |
| *TRIM15* | Regulation of innate immunity (likely) | 2 | 2.26 | 0.630 | 0.00076965 |
| *APOBEC1* | RNA editing, C-to-U deamination | 2 | 2.24 | 0.571 | 0.02259335 |
| *XKR9* | Unknown function. Predicted protein. | 2 | 2.23 | 0.226 | 0.00001518 |

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|---|---|---|---|---|---|
| *VAX2* | Transcription factor, eye and forebrain development | 2 | 2.22 | 0.569 | **0.00000034** |
| *OR2A1* | Olfactory signaling | 3 | 2.21 | 0.817 | 0.18544589 |
| *GRK1* | Phototransduction, receptor desensitization | 2 | 2.18 | 0.775 | 0.00028900 |
| *LRIT3* | Retinal bipolar cell function | 2 | 2.18 | 0.464 | **0.00000000** |
| *OR10J1* | Olfactory signaling | 2 | 2.18 | 0.289 | **0.00000000** |
| *TET2* | DNA demethylation, hematopoietic regulation | 2 | 2.18 | 1.047 | 0.65911065 |
| *POU4F1* | Transcription factor, hair cell development | 3 | 2.16 | 0.576 | 0.00031832 |
| *GABRG3* | GABAergic neurotransmission, inhibitory neurotransmission | 2 | 2.14 | 0.611 | **0.00000011** |
| *CXCL10* | Chemotaxis, immune cell recruitment | 2 | 2.14 | 0.271 | 0.03261090 |
| *GRM6* | Glutamate signaling in ON bipolar cells, retinal function | 3 | 2.13 | 0.734 | **0.00000000** |
| *RDH5* | Retinoid metabolism, visual cycle | 2 | 2.12 | 0.572 | **0.00000410** |
| *ACTRT1* | Actin-related protein. Function unknown. | 2 | 2.12 | 0.583 | **0.00000638** |
| *OR5D14* | Olfactory signaling | 2 | 2.11 | 1.220 | 0.00980333 |
| *FAM26F* | Immune response, regulation of inflammation | 3 | 2.11 | 0.491 | 0.02365873 |
| *DNM2* | Endocytosis, vesicle trafficking | 2 | 2.10 | 0.728 | **0.00000094** |
| *OR1L6* | Olfactory signaling | 2 | 2.10 | 0.849 | 0.16748881 |
| *NOL10* | Ribosome biogenesis, nucleolar protein | 2 | 2.09 | 0.915 | 0.20115012 |
| *C2orf72* | Ciliary function, photoreceptor maintenance (potentially) | 2 | 2.08 | 0.916 | 0.56214940 |
| *PGLYRP3* | Peptidoglycan recognition, innate immunity | 2 | 2.07 | 0.742 | 0.08694424 |
| *TAAR3* | Trace amine-associated receptor, olfactory signaling/behavior (potentially) | 3 | 2.07 | 0.311 | **0.00000000** |
| *GPR55* | Cannabinoid receptor (atypical), signaling in various tissues | 3 | 2.06 | 0.295 | **0.00000002** |
| *CASP7* | Apoptosis executioner caspase | 2 | 2.05 | 0.000 | **0.00000018** |
| *IFNA6* | Type I interferon, antiviral response | 2 | 2.05 | 1.223 | 0.41335071 |
| *GJA9* | Gap junction communication, lens development | 2 | 2.05 | 0.000 | **0.00000043** |
| *ZNF649* | Transcription regulation (likely) | 3 | 2.05 | 0.052 | 0.00007690 |
| *OR9K2* | Olfactory signaling | 3 | 2.04 | 1.045 | 0.77931981 |
| *GSDMB* | Gasdermin family, pyroptosis (potentially) | 2 | 2.04 | 1.712 | 0.00356124 |
| *AP1AR* | Vesicle trafficking, protein sorting | 2 | 2.04 | 0.849 | 0.08667165 |
| *OR52B6* | Olfactory signaling | 3 | 2.04 | 0.950 | 0.67499098 |
| *MYO10* | Unconventional myosin, intracellular transport, filopodia formation | 3 | 2.04 | 0.767 | **0.00000000** |
| *GFAP* | Astrocyte intermediate filament, glial cell marker | 2 | 2.03 | 0.724 | 0.00002967 |
| *HBBP1* | Heme binding, pseudogene of hemoglobin. | 2 | 2.03 | 0.580 | 0.00501897 |
| *RAB19* | Vesicle trafficking, membrane transport | 2 | 2.02 | 0.827 | 0.12741018 |
| *ZNF217* | Transcription factor, oncogene, chromatin remodeling | 2 | 2.02 | 1.090 | 0.10950123 |
| *OR52H1* | Olfactory signaling | 2 | 2.01 | 1.049 | 0.62451780 |
| *CASP6* | Apoptosis executioner caspase | 2 | 2.01 | 0.866 | 0.25655505 |
| *ZNF615* | Transcription regulation (likely) | 3 | 2.00 | 0.903 | 0.38030723 |

**Table S1.** Genes with accelerated rates on the subterranean lineages (≥2 ratio), with FDR ≤ 0.01 for the Proportional: Proportional Partitioned test and at least 2 informative branches. The genes are sorted on the estimated rate ratio (decreasing). RELAX relaxation/intensification parameters K (color coded for K<1 or K>1) and p-values (color coded for p ≤ 0.00001) are reported for comparing subterranean branches with the rest of the tree. Gene process annotation automatically generated by `Gemini 2.5 Pro experimental`.

| Gene ID | Process | Branch Count | Relative Rate | RELAX K | RELAX p-value |
|---|---|---|---|---|---|
| *TRIM27* | Ubiquitination, transcriptional regulation, immune response | 2 | 0.00 | 14.32 | 0.0000020 |
| *SP9* | Transcription factor, skeletal development, limb development | 3 | 0.00 | 50.00 | 0.0032560 |
| *NOVA2* | RNA-binding protein, alternative splicing regulation (nervous system) | 3 | 0.00 | 21.44 | 0.0000669 |
| *SPRR2E* | Keratinocyte differentiation, skin barrier formation | 2 | 0.05 | 2.31 | 0.0655754 |
| *SPRR2A* | Keratinocyte differentiation, skin barrier formation | 2 | 0.06 | 26.32 | 0.0253551 |
| *SPRR2B* | Keratinocyte differentiation, skin barrier formation | 2 | 0.06 | 29.17 | 0.0025212 |
| *NSFL1C* | Cofactor for p97 ATPase, protein degradation | 3 | 0.09 | 1.84 | 0.0000076 |
| *PURG* | RNA helicase activity, regulation of transcription from RNA polymerase II promoter. | 3 | 0.10 | 1.77 | 0.0006125 |
| *TSSK2* | Spermatogenesis, serine/threonine kinase activity | 2 | 0.16 | 1.85 | 0.0000060 |
| *FZD8* | Wnt signaling pathway receptor | 2 | 0.17 | 1.92 | 0.0000452 |
| *RNF187* | E3 ubiquitin-protein ligase activity | 2 | 0.18 | 1.66 | 0.0382464 |
| *SP8* | Transcription factor, limb and brain development | 3 | 0.19 | 8.01 | 0.0002783 |
| *OR2T11* | Olfactory receptor | 2 | 0.22 | 2.08 | 0.0016966 |
| *ONECUT3* | Transcription factor, liver development, pancreas development | 2 | 0.22 | 1.68 | 0.0000463 |
| *USP9Y* | Deubiquitinating enzyme, spermatogenesis | 3 | 0.24 | 1.28 | 0.0000001 |
| *KCNA7* | Potassium channel, ion transport | 3 | 0.25 | 2.66 | 0.0609791 |
| *CALML3* | Calcium-binding protein, skin development | 2 | 0.26 | 1.46 | 0.0071034 |
| *AASDHPPT* | Amino acid metabolism | 2 | 0.26 | 12.55 | 0.0000213 |
| *HBD* | Beta-globin chain of hemoglobin | 2 | 0.27 | 2.62 | 0.0021488 |
| *ZNF189* | Transcription factor | 3 | 0.27 | 1.52 | 0.0002477 |
| *OR5T2* | Olfactory receptor | 3 | 0.28 | 1.63 | 0.0007659 |
| *OR4K17* | Olfactory receptor | 3 | 0.28 | 1.03 | 0.8682111 |
| *FOXD3* | Transcription factor, neural crest development | 2 | 0.28 | 1.26 | 0.0103134 |
| *OR3A4P* | Olfactory receptor (pseudogene) | 2 | 0.28 | 1.33 | 0.4268307 |
| *OR51G1* | Olfactory receptor | 3 | 0.29 | 1.51 | 0.0000773 |
| *CT45A6* | Cancer/testis antigen | 2 | 0.30 | 4.14 | 0.0007862 |
| *UTY* | Histone demethylase, Y chromosome homolog of UTX | 3 | 0.30 | 1.42 | 0.0002247 |
| *CT45A5* | Cancer/testis antigen | 2 | 0.31 | 5.11 | 0.0012796 |
| *DHX40* | RNA helicase activity. | 2 | 0.31 | 1.78 | 0.0003389 |
| *OR1G1* | Olfactory receptor | 2 | 0.31 | 1.44 | 0.0119643 |
| *SCRIB* | Cell polarity, cell adhesion | 2 | 0.31 | 1.01 | 0.8956428 |
| *SLC39A3* | Zinc transporter | 2 | 0.32 | 1.42 | 0.0007238 |
| *OR5K2* | Olfactory receptor | 2 | 0.32 | 4.40 | 0.0001178 |
| *OR2Y1* | Olfactory receptor | 2 | 0.33 | 1.07 | 0.7216611 |
| *TOR1A* | Protein folding, ER stress response | 2 | 0.33 | 1.36 | 0.0007150 |
| *ZNF260* | Transcription factor | 2 | 0.33 | 1.52 | 0.0047839 |
| *PRPS1L1* | Ribose-phosphate pyrophosphokinase activity (purine synthesis) | 2 | 0.33 | 1.39 | 0.0004886 |

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|---|---|---|---|---|---|
| *PRSS3P2* | Serine protease (pseudogene) | 2 | 0.34 | 4.67 | 0.0009085 |
| *ZFP3* | Transcription factor | 3 | 0.34 | 1.55 | 0.0010914 |
| *OR1B1* | Olfactory receptor | 2 | 0.34 | 1.11 | 0.7043079 |
| *CITED1* | Transcriptional coactivator | 2 | 0.34 | 2.04 | 0.0334987 |
| *PCTP* | Phosphatidylcholine transfer | 3 | 0.35 | 21.96 | 0.0086924 |
| *PPFIA3* | Regulation of cell adhesion and migration | 2 | 0.35 | 1.18 | 0.0369161 |
| *C14orf169* | Protein of unknown function. | 2 | 0.35 | 1.34 | 0.0000518 |
| *SLC38A1* | Amino acid transport | 2 | 0.35 | 2.13 | 0.0000002 |
| *LRRC41* | Leucine-rich repeat-containing protein, function unknown. | 3 | 0.35 | 1.37 | 0.0014356 |
| *SCD5* | Fatty acid desaturation | 2 | 0.36 | 1.39 | 0.1044302 |
| *OR4F4* | Olfactory receptor | 3 | 0.36 | 1.36 | 0.0438351 |
| *NUF2* | Kinetochore complex component, chromosome segregation | 2 | 0.36 | 1.47 | 0.0079767 |
| *GLI4* | Transcription factor, Hedgehog signaling pathway | 2 | 0.37 | 2.06 | 0.0087036 |
| *OR1J4* | Olfactory receptor | 3 | 0.37 | 1.77 | 0.0000940 |
| *PEG10* | Retrotransposon-derived, cell proliferation | 2 | 0.37 | 1.20 | 0.0874201 |
| *TRIM11* | Ubiquitination, innate immunity | 2 | 0.37 | 1.27 | 0.0346460 |
| *OR5M10* | Olfactory receptor | 3 | 0.37 | 1.23 | 0.0734114 |
| *GNRHR2* | Gonadotropin-releasing hormone receptor | 2 | 0.37 | 3.88 | 0.0000475 |
| *TAAR5* | Trace amine-associated receptor | 2 | 0.38 | 1.77 | 0.0666801 |
| *OR1N1* | Olfactory receptor | 2 | 0.38 | 1.43 | 0.0052453 |
| *ASCL3* | Transcription factor, neurogenesis | 2 | 0.38 | 1.26 | 0.2108113 |
| *OR51L1* | Olfactory receptor | 3 | 0.38 | 1.27 | 0.1271770 |
| *OR6C75* | Olfactory receptor | 2 | 0.38 | 1.14 | 0.2966716 |
| *OR5AS1* | Olfactory receptor | 3 | 0.38 | 1.08 | 0.4849471 |
| *GPR75* | G-protein coupled receptor, signal transduction | 2 | 0.38 | 1.23 | 0.0142863 |
| *ALDH3B2* | Aldehyde dehydrogenase | 2 | 0.39 | 1.50 | 0.0031803 |
| *OR51G2* | Olfactory receptor | 2 | 0.39 | 1.25 | 0.1039413 |
| *NDC80* | Kinetochore complex component, chromosome segregation | 2 | 0.40 | 1.35 | 0.0004120 |
| *OR8G2* | Olfactory Receptor | 2 | 0.40 | 1.03 | 0.8862974 |
| *LINC00282* | Long non-coding RNA, function largely unknown. | 2 | 0.40 | 2.06 | 0.0014156 |
| *MYT1L* | Myelin transcription factor 1-like, neuronal development | 2 | 0.41 | 1.08 | 0.4859425 |
| *FOXD2* | Transcription regulator activity. | 2 | 0.41 | 1.63 | 0.0277729 |
| *C1QBP* | Regulation of complement activation, inflammation, and transcription. | 2 | 0.41 | 1.64 | 0.0113296 |
| *ECE1* | Endothelin-converting enzyme, peptide hormone processing | 2 | 0.41 | 1.21 | 0.0182194 |
| *SLC22A11* | Organic anion transporter | 2 | 0.42 | 1.80 | 0.0103551 |
| *OR6Y1* | Olfactory receptor | 2 | 0.42 | 1.23 | 0.1679264 |
| *ZNF281* | Transcriptional repressor activity, DNA-binding. | 2 | 0.42 | 0.98 | 0.8823412 |
| *CYP1A2* | Cytochrome P450, xenobiotic metabolism | 2 | 0.42 | 1.56 | 0.0053253 |
| *ZNF74* | Zinc finger protein involved in transcriptional regulation. | 2 | 0.42 | 1.05 | 0.6833111 |
| *OR4F17* | Olfactory receptor | 2 | 0.42 | 1.41 | 0.0161296 |

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|---|---|---|---|---|---|
| *ZSWIM5* | Involved in protein ubiquitination. | 3 | 0.42 | 1.16 | 0.2161543 |
| *PLXNA1* | Involved in axon guidance and cell migration. | 2 | 0.42 | 1.35 | 0.0000050 |
| *ZNF410* | Transcription factor. | 2 | 0.42 | 1.40 | 0.0032347 |
| *RNF175* | Involved in protein ubiquitination | 2 | 0.42 | 3.01 | 0.0042804 |
| *ZFP82* | Transcription factor. | 3 | 0.43 | 1.33 | 0.0019230 |
| *ZNF689* | Transcription factor involved in cell differentiation. | 2 | 0.44 | 1.05 | 0.6625125 |
| *HRH2* | Histamine H2 receptor, involved in gastric acid secretion. | 2 | 0.44 | 1.11 | 0.5791339 |
| *OR6C2* | Olfactory receptor | 2 | 0.44 | 1.09 | 0.4253905 |
| *ZNF221* | Transcription factor. | 3 | 0.44 | 46.21 | 0.0004550 |
| *ACOT13* | Acyl-CoA thioesterase, lipid metabolism | 2 | 0.44 | 1.84 | 0.6624897 |
| *GPR161* | G protein-coupled receptor, involved in cAMP signaling. | 2 | 0.44 | 1.23 | 0.0073362 |
| *ZNF567* | Transcription factor. | 3 | 0.44 | 1.41 | 0.0104547 |
| *RNASE1* | Ribonuclease, RNA degradation | 2 | 0.44 | 2.16 | 0.0175659 |
| *DHRS12* | Short-chain dehydrogenase/reductase, involved in retinoid metabolism. | 2 | 0.44 | 1.37 | 0.2337361 |
| *ZNF3* | Transcription factor | 2 | 0.44 | 1.43 | 0.0603739 |
| *OR51I2* | Olfactory receptor | 2 | 0.44 | 1.29 | 0.0127354 |
| *OR8D4* | Olfactory receptor | 3 | 0.44 | 1.34 | 0.0285834 |
| *ALOX15* | Arachidonate 15-lipoxygenase, lipid metabolism | 2 | 0.45 | 1.29 | 0.1363433 |
| *OR4K14* | Olfactory receptor | 3 | 0.45 | 1.41 | 0.0268919 |
| *OR1L4* | Olfactory receptor | 2 | 0.45 | 1.40 | 0.0213765 |
| *OR2D3* | Olfactory receptor | 2 | 0.45 | 1.74 | 0.0131097 |
| *OR5H1* | Olfactory receptor | 3 | 0.45 | 2.08 | 0.0001634 |
| *KDM5B* | Histone demethylase, transcriptional regulation | 3 | 0.45 | 1.23 | 0.0180806 |
| *OR5L1* | Olfactory receptor | 3 | 0.45 | 1.21 | 0.1753852 |
| *SHANK1* | Scaffolding protein in postsynaptic density | 3 | 0.45 | 1.09 | 0.0603943 |
| *PREPL* | Protease, involved in protein processing | 2 | 0.46 | 1.11 | 0.3580791 |
| *C6orf118* | Protein of unknown function | 2 | 0.46 | 1.00 | 1.0000000 |
| *HENMT1* | Small RNA methyltransferase activity | 3 | 0.46 | 1.24 | 0.1167612 |
| *RNF214* | E3 ubiquitin-protein ligase activity | 2 | 0.46 | 1.57 | 0.0004362 |
| *ZFP1* | Transcription factor involved in developmental processes | 2 | 0.46 | 1.03 | 0.8340600 |
| *ZBTB9* | Transcription regulator activity, DNA-binding | 2 | 0.47 | 1.38 | 0.0034090 |
| *GRIN3B* | Glutamate receptor subunit, synaptic transmission | 2 | 0.47 | 1.32 | 0.0000799 |
| *KBTBD6* | Substrate-specific adapter of a BCR (BTB-CUL3-RBX1) E3 ubiquitin-protein ligase complex. | 2 | 0.47 | 1.41 | 0.0001256 |
| *OR5V1* | Olfactory receptor | 2 | 0.47 | 1.52 | 0.0124883 |
| *GAB4* | Signal transduction, adapter protein | 2 | 0.47 | 1.29 | 0.0099495 |
| *OR1J1* | Olfactory receptor | 2 | 0.47 | 1.24 | 0.2329807 |
| *OR1Q1* | Olfactory receptor | 2 | 0.48 | 1.11 | 0.4664150 |
| *UPP2* | Uridine phosphorylase activity | 2 | 0.48 | 1.30 | 0.0583165 |
| *REN* | Renin, blood pressure regulation | 2 | 0.48 | 1.76 | 0.0001486 |
| *OR4F5* | Olfactory receptor | 2 | 0.48 | 1.32 | 0.0345714 |

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|---|---|---|---|---|---|
| *SCN5A* | Voltage-gated sodium channel, cardiac muscle contraction | 3 | 0.48 | 1.20 | 0.0038741 |
| *P2RY2* | Purinergic receptor, G-protein coupled | 2 | 0.48 | 1.13 | 0.3420303 |
| *SLC17A2* | Urate transport/absorption | 3 | 0.48 | 1.29 | 0.0631097 |
| *OR9G9* | Olfactory receptor | 3 | 0.48 | 1.19 | 0.1790358 |
| *OR1A1* | Olfactory receptor | 2 | 0.48 | 1.14 | 0.4028322 |
| *GSTA5* | Glutathione S-transferase, detoxification | 2 | 0.49 | 8.52 | 0.0081953 |
| *ZNF530* | Transcription factor | 2 | 0.49 | 1.04 | 0.8422095 |
| *DARC* | Atypical chemokine receptor, chemokine binding | 2 | 0.49 | 1.48 | 0.0606461 |
| *EPM2AIP1* | Regulation of glycogen metabolism | 2 | 0.49 | 0.98 | 0.8960868 |
| *SFMBT2* | Transcriptional Repression | 2 | 0.49 | 1.17 | 0.2041710 |
| *ZNF287* | Transcription factor | 2 | 0.50 | 1.21 | 0.0757646 |
| *CACNA1E* | Voltage-dependent calcium channel, involved in neurotransmitter release | 2 | 0.50 | 1.31 | 0.0000040 |
| *SEC14L4* | Phospholipid binding | 2 | 0.50 | 1.25 | 0.1139909 |
| *ADARB2* | RNA editing | 2 | 0.50 | 1.11 | 0.2112145 |

**Table S2.** Genes with decelerated rates on the subterranean lineages (≥2 ratio), with FDR ≤ 0.01 for the Proportional: Proportional Partitioned test and at least 2 informative branches. The genes are sorted on the estimated rate ratio (increasing). RELAX relaxation/intensification parameters K (color coded for K<1 or K>1) and p-values (color coded for p ≤ 0.00001) are reported for comparing subterranean branches with the rest of the tree. Gene process annotation automatically generated by `Gemini 2.5 Pro experimental`.

| Gene ID | Process | Branch Count | Relative Rate | RELAX K | RELAX p-value |
|---|---|---|---|---|---|
| *OR51G2* | Olfaction (sensory perception of smell) | 3 | 23.33 | 0.65 | 0.001938 |
| *OR51G1* | Olfaction (sensory perception of smell) | 3 | 16.84 | 0.54 | 0.000000 |
| *OR51I2* | Olfaction (sensory perception of smell) | 3 | 13.65 | 0.59 | 0.000001 |
| *OR4K14* | Olfaction (sensory perception of smell) | 3 | 11.47 | 0.40 | 0.000000 |
| *OR51I1* | Olfaction (sensory perception of smell) | 3 | 10.95 | 0.52 | 0.000000 |
| *KRTAP9-8* | Hair keratin formation | 3 | 10.43 | 0.02 | 0.000039 |
| *OR52E6* | Olfaction (sensory perception of smell) | 3 | 8.90 | 1.02 | 0.820461 |
| *OR2V1* | Olfaction (sensory perception of smell) | 3 | 7.97 | 0.42 | 0.000307 |
| *RNF187* | Protein ubiquitination | 3 | 7.22 | 0.70 | 0.030494 |
| *CTAG1B* | Cancer/testis antigen (immune response) | 3 | 7.21 | 0.00 | 0.024873 |
| *OR4S1* | Olfaction (sensory perception of smell) | 3 | 7.00 | 1.18 | 0.026798 |
| *MYH13* | Muscle contraction | 5 | 6.89 | 0.18 | 0.000000 |
| *OR56A5* | Olfaction (sensory perception of smell) | 3 | 6.80 | 0.74 | 0.003953 |
| *ZNF791* | Transcription regulation | 3 | 6.78 | 0.57 | 0.000001 |
| *OR51L1* | Olfaction (sensory perception of smell) | 3 | 6.76 | 0.51 | 0.000276 |
| *OR51B2* | Olfaction (sensory perception of smell) | 3 | 6.70 | 1.03 | 0.678273 |
| *OR51A7* | Olfaction (sensory perception of smell) | 4 | 6.53 | 0.76 | 0.001256 |
| *OR4C11* | Olfaction (sensory perception of smell) | 3 | 6.33 | 0.77 | 0.005954 |
| *OR51Q1* | Olfaction (sensory perception of smell) | 3 | 6.19 | 0.83 | 0.068506 |
| *MYH4* | Muscle contraction | 4 | 6.15 | 0.36 | 0.000000 |
| *ZNF749* | Transcription regulation | 4 | 5.83 | 1.02 | 0.879015 |

| | | | | | |
|---|---|---|---|---|---|
| *OR6K6* | Olfaction (sensory perception of smell) | 3 | 5.75 | 0.84 | 0.088249 |
| *OR12D2* | Olfaction (sensory perception of smell) | 3 | 5.67 | 0.79 | 0.084178 |
| *OR3A1* | Olfaction (sensory perception of smell) | 3 | 5.34 | 0.55 | 0.000363 |
| *OR8B3* | Olfaction (sensory perception of smell) | 3 | 5.12 | 0.90 | 0.248547 |
| *MAGEA4* | Cancer/testis antigen (immune response, may play a role in tumor progression) | 4 | 4.79 | 0.83 | 0.364459 |
| *KRTAP10-8* | Hair keratin formation | 3 | 4.78 | 0.76 | 0.093999 |
| *HLA-C* | Immune response (antigen presentation) | 4 | 4.53 | 0.00 | 0.015887 |
| *ZNF17* | Transcription regulation | 3 | 4.50 | 1.10 | 0.269954 |
| *ANKRD23* | Muscle development and function; sarcomere organization | 3 | 4.40 | 0.12 | 0.000000 |
| *UPF1* | RNA surveillance (nonsense-mediated mRNA decay) | 3 | 4.40 | 0.60 | 0.000000 |
| *ZNF709* | Transcription regulation | 3 | 4.39 | 0.47 | 0.000000 |
| *OR6C6* | Olfaction (sensory perception of smell) | 3 | 4.30 | 1.00 | 1.000000 |
| *ZNF568* | Transcription regulation | 5 | 4.27 | 0.75 | 0.004226 |
| *GNAT3* | G-protein signaling (taste transduction) | 3 | 4.25 | 0.08 | 0.000000 |
| *LEP* | Regulation of body weight, metabolism, and appetite | 3 | 4.18 | 0.00 | 0.000000 |
| *OR8S1* | Olfaction (sensory perception of smell) | 4 | 4.10 | 0.72 | 0.000083 |
| *SLC47A1* | Transporter (drug and toxin excretion) | 4 | 3.97 | 0.00 | 0.000000 |
| *NOX5* | Reactive oxygen species (ROS) generation | 4 | 3.97 | 0.13 | 0.000000 |
| *OR5B2* | Olfaction (sensory perception of smell) | 3 | 3.94 | 0.91 | 0.241300 |
| *UTY* | Histone demethylation (male-specific) | 3 | 3.87 | 0.81 | 0.003775 |
| *AX747376* | Unknown Function | 4 | 3.75 | 0.87 | 0.502306 |
| *ZNF547* | Transcription regulation | 4 | 3.71 | 0.74 | 0.124231 |
| *ZNF569* | Transcription regulation | 4 | 3.62 | 0.41 | 0.000000 |
| *KRTAP10-12* | Hair keratin formation | 3 | 3.59 | 0.36 | 0.138234 |
| *OR2A2* | Olfaction (sensory perception of smell) | 3 | 3.58 | 0.75 | 0.002997 |
| *ZNF12* | Transcription regulation | 4 | 3.51 | 0.77 | 0.016043 |
| *GSTM4* | Detoxification of xenobiotics | 3 | 3.48 | 0.19 | 0.000899 |
| *OR6B1* | Olfaction (sensory perception of smell) | 3 | 3.38 | 0.23 | 0.000000 |
| *ZNF136* | Transcription regulation | 3 | 3.35 | 0.94 | 0.754785 |
| *SSTR4* | Somatostatin receptor (hormone signaling) | 5 | 3.34 | 0.21 | 0.000000 |
| *MYH1* | Muscle contraction | 5 | 3.33 | 0.16 | 0.000000 |
| *SLC2A4* | Glucose transport | 3 | 3.27 | 0.47 | 0.000000 |
| *GCKR* | Glucokinase regulator (glucose metabolism) | 3 | 3.19 | 0.00 | 0.000000 |
| *TCRBV5S1A1T* | T-cell receptor (immune response) | 3 | 3.15 | 0.02 | 0.044341 |
| *NT5DC4* | 5'-nucleotidase activity | 3 | 3.13 | 13.67 | 0.000000 |
| *OR5H2* | Olfaction (sensory perception of smell) | 3 | 3.03 | 0.53 | 0.000225 |
| *KRT74* | Keratin formation (epithelial cells) | 3 | 3.03 | 0.62 | 0.000198 |
| *CYP2F1* | Xenobiotic metabolism | 3 | 3.02 | 0.48 | 0.000000 |
| *OR10G3* | Olfaction (sensory perception of smell) | 3 | 2.98 | 0.27 | 0.000000 |
| *ZNF671* | Transcription regulation | 4 | 2.97 | 0.71 | 0.139460 |
| *CD46* | Complement regulation | 4 | 2.95 | 1.22 | 0.056122 |

| | | | | | |
|---|---|---|---|---|---|
| *TGM3* | Protein cross-linking (epidermis development) | 4 | 2.94 | 0.03 | 0.000000 |
| *OR10S1* | Olfaction (sensory perception of smell) | 3 | 2.92 | 0.03 | 0.000000 |
| *EPPK1* | Cell envelope structural component | 4 | 2.88 | 0.72 | 0.000000 |
| *ZNF324B* | Transcription regulation | 3 | 2.86 | 0.00 | 0.000000 |
| *ATP2A1* | Calcium transport (muscle contraction/relaxation) | 5 | 2.84 | 0.57 | 0.000000 |
| *ZNF552* | Transcription regulation | 3 | 2.82 | 0.61 | 0.157571 |
| *ZNF835* | Transcription regulation | 3 | 2.77 | 1.62 | 0.000000 |
| *ZNF134* | Transcription regulation | 4 | 2.77 | 0.48 | 0.000758 |
| *ALDH3B2* | Detoxification | 3 | 2.76 | 0.00 | 0.000001 |
| *GRIN3B* | Glutamate receptor | 5 | 2.76 | 0.35 | 0.000000 |
| *PRSS3P2* | Serine protease | 3 | 2.73 | 0.55 | 0.515155 |
| *ZNF426* | Transcription regulation | 3 | 2.73 | 0.81 | 0.297724 |
| *CHIA* | Chitinase activity | 4 | 2.73 | 0.00 | 0.000000 |
| *AQP10* | Water and small solute transport | 3 | 2.72 | 0.17 | 0.000018 |
| *CYP3A43* | Xenobiotic metabolism | 4 | 2.72 | 0.19 | 0.000007 |
| *CYP2S1* | Xenobiotic metabolism | 3 | 2.70 | 0.41 | 0.000004 |
| *ZNF585B* | Transcription regulation | 3 | 2.66 | 0.00 | 0.000000 |
| *ZNF551* | Transcription regulation | 3 | 2.66 | 0.73 | 0.090825 |
| *KRTAP10-3* | Hair keratin formation | 5 | 2.64 | 0.83 | 0.361442 |
| *COL9A2* | Collagen fibril organization | 3 | 2.64 | 0.37 | 0.007256 |
| *ZSCAN31* | Transcription regulation | 5 | 2.59 | 0.12 | 0.001741 |
| *LILRA1* | Immune response | 3 | 2.57 | 1.16 | 0.405857 |
| *MAZ* | Transcription regulation | 3 | 2.53 | 0.19 | 0.000000 |
| *OTUD6A* | Deubiquitinase activity | 3 | 2.48 | 0.75 | 0.427064 |
| *OR51D1* | Olfaction (sensory perception of smell) | 3 | 2.48 | 0.28 | 0.000000 |
| *SBSN* | Cell adhesion | 4 | 2.47 | 1.51 | 0.002276 |
| *MYBPC1* | Muscle contraction | 3 | 2.46 | 0.00 | 0.000000 |
| *TERF2IP* | Telomere maintenance and DNA damage response | 4 | 2.46 | 1.22 | 0.286071 |
| *OR4L1* | Olfaction (sensory perception of smell) | 3 | 2.44 | 0.13 | 0.000462 |
| *NPSR1* | Neuropeptide S receptor | 3 | 2.42 | 0.22 | 0.000000 |
| *OR2H2* | Olfaction (sensory perception of smell) | 3 | 2.41 | 0.00 | 0.000000 |
| *ZNF416* | Transcription regulation | 5 | 2.39 | 1.42 | 0.000384 |
| *ZNF260* | Transcription regulation | 3 | 2.39 | 0.70 | 0.002775 |
| *PAPL* | Protease activity | 3 | 2.37 | 0.47 | 0.000000 |
| *FER1L4* | Membrane fusion and calcium signaling | 3 | 2.36 | 0.13 | 0.000000 |
| *TAAR9* | Trace amine-associated receptor (olfaction) | 3 | 2.35 | 0.00 | 0.000000 |
| *ZNF548* | Transcription regulation | 4 | 2.33 | 1.46 | 0.079499 |
| *OR52M1* | Olfaction (sensory perception of smell) | 3 | 2.31 | 0.54 | 0.000038 |
| *NKD2* | Wnt signaling pathway antagonist | 3 | 2.29 | 0.63 | 0.000754 |
| *ZNF560* | Transcription regulation | 4 | 2.28 | 0.66 | 0.066371 |
| *ZNF665* | Transcription regulation | 4 | 2.27 | 0.67 | 0.007559 |
| *NLRP4* | Inflammasome formation and innate immunity | 3 | 2.25 | 0.61 | 0.006940 |

| | | | | | |
|---|---|---|---|---|---|
| *ULK2* | Autophagy | 3 | 2.21 | 0.58 | 0.010387 |
| *CAPN8* | Calcium-activated protease | 3 | 2.21 | 0.43 | 0.000326 |
| *OTOF* | Synaptic vesicle fusion (hearing) | 4 | 2.20 | 0.72 | 0.000001 |
| *ZSCAN5A* | Transcription regulation | 3 | 2.19 | 0.69 | 0.592246 |
| *ZNF224* | Transcription regulation | 3 | 2.18 | 0.70 | 0.000603 |
| *ZEB1* | Epithelial-mesenchymal transition (EMT), transcription factor | 3 | 2.18 | 0.74 | 0.000129 |
| *MAGEA10* | Cancer/testis antigen | 3 | 2.16 | 0.22 | 0.017760 |
| *ZNF812* | Transcription regulation | 3 | 2.16 | 0.61 | 0.061390 |
| *ZNF684* | Transcription regulation | 3 | 2.16 | 0.68 | 0.134173 |
| *NEB* | Muscle actin filament organization | 3 | 2.14 | 0.32 | 0.000000 |
| *PRKAG3* | AMPK gamma subunit | 3 | 2.11 | 0.00 | 0.000000 |
| *ZNF766* | Transcription regulation | 3 | 2.10 | 0.43 | 0.000288 |
| *FMO2* | Xenobiotic metabolism | 3 | 2.06 | 0.24 | 0.000000 |
| *ZNF345* | Transcription regulation | 5 | 2.04 | 0.69 | 0.000173 |
| *ZNF540* | Transcription regulation | 4 | 2.04 | 1.03 | 0.796812 |
| *ZNF829* | Transcription regulation | 3 | 2.03 | 1.31 | 0.377747 |

**Table S3.** Genes with accelerated rates on the marine lineages (≥2 ratio), with FDR ≤ 0.01 for the Proportional: Proportional Partitioned test and at least 3 informative branches. The genes are sorted on the estimated rate ratio (decreasing). RELAX relaxation/intensification parameters K (color coded for K<1 or K>1) and p-values (color coded for p ≤ 0.00001) are reported for comparing subterranean branches with the rest of the tree. Gene process annotation automatically generated by `Gemini 2.5 Pro experimental`.

| Gene ID | Process | Branch Count | Relative Rate | RELAX K | RELAX p-value |
|---|---|---|---|---|---|
| *ZNF585A* | Transcription regulation | 3 | 0.00 | 3.6 | 0.005828457 |
| *CXorf57* | Unknown, but likely involved in male fertility | 4 | 0.05 | 9.6 | 0.000000010 |
| *NANOS1* | Germ cell development, translational regulation | 3 | 0.11 | 0.9 | 0.621168070 |
| *TRIM61* | Innate immunity, viral restriction | 3 | 0.11 | 1.3 | 0.622878259 |
| *KCNC3* | Voltage-gated potassium channel activity, neuronal excitability | 3 | 0.12 | 1.7 | 0.423506345 |
| *ZNF331* | Transcription regulation | 4 | 0.12 | 1.7 | 0.146956847 |
| *TRIM11* | Ubiquitin ligase activity, protein degradation | 3 | 0.15 | 50.0 | 0.024677239 |
| *PRDM7* | Histone methylation, transcriptional regulation | 3 | 0.16 | 1.4 | 0.710043451 |
| *LOC93622* | Unknown Function | 3 | 0.16 | 1.1 | 0.750257856 |
| *BEGAIN* | Brain development, neurite outgrowth | 3 | 0.17 | 1.2 | 0.073258594 |
| *ZNF890P* | Likely a pseudogene, so likely non-functional. | 3 | 0.19 | 6.8 | 0.021915985 |
| *SHISA8* | regulation of AMPA receptor trafficking | 3 | 0.19 | 0.2 | 0.000183612 |
| *ADAM29* | Proteolysis, cell adhesion, metalloproteinase activity | 4 | 0.19 | 1.0 | 0.915639665 |
| *LAS1L* | Ribosome biogenesis | 5 | 0.20 | 38.1 | 0.000449155 |
| *SLC38A5* | Amino acid transport | 3 | 0.22 | 1.7 | 0.042476091 |
| *PEG3* | Imprinted gene, involved in cell growth, apoptosis | 5 | 0.22 | 1.6 | 0.009011513 |
| *GDF7* | Growth factor, bone and cartilage development | 3 | 0.22 | 1.2 | 0.412940082 |
| *ZNF75D* | Transcription regulation | 4 | 0.23 | 2.8 | 0.049479880 |
| *BTNL2* | Immune regulation, T cell activation modulation | 3 | 0.24 | 2.1 | 0.472628988 |

| | | | | | |
|---|---|---|---|---|---|
| *CNST* | Centrosome and spindle organization, likely cell cycle | 4 | 0.26 | 0.3 | 0.002103799 |
| *MAGEL2* | Imprinted gene, involved in circadian rhythm, neuronal development | 5 | 0.26 | 1.7 | 0.001499586 |
| *ZNF490* | Transcription regulation | 3 | 0.26 | 0.9 | 0.846568002 |
| *ADCY10* | Adenylate cyclase activity, cAMP synthesis | 5 | 0.26 | 2.0 | 0.150779858 |
| *RBM12B* | RNA binding | 4 | 0.27 | 1.4 | 0.132693210 |
| *ZNF133* | Transcription regulation | 4 | 0.27 | 1.2 | 0.319362678 |
| *CXXC11* | Unknown Function | 5 | 0.27 | 1.3 | 0.184679581 |
| *ZNF527* | Transcription regulation | 4 | 0.28 | 8.6 | 0.021161902 |
| *IFIT5* | Antiviral defense, RNA binding | 4 | 0.29 | 1.5 | 0.101261346 |
| *SLC9C1* | Sodium-hydrogen exchange | 5 | 0.30 | 1.4 | 0.198522562 |
| *NLRP8* | Innate immunity, inflammasome formation | 4 | 0.30 | 1.3 | 0.267213358 |
| *TIGD6* | Unknown, Tigger transposable element derived 6 | 4 | 0.31 | 1.2 | 0.548797249 |
| *SIGLECL1* | Unknown, but likely immune cell regulation | 3 | 0.31 | 0.0 | 0.432445779 |
| *SPATA13* | Spermatogenesis | 5 | 0.31 | 0.8 | 0.050926030 |
| *ZFP82* | Transcription regulation | 3 | 0.31 | 1.2 | 0.173540359 |
| *CYP11B1* | Steroid hormone biosynthesis (cortisol synthesis) | 3 | 0.31 | 1.0 | 1.000000000 |
| *PRSS55* | Serine protease, function largely unknown | 3 | 0.32 | 1.1 | 0.756180108 |
| *ZNF317* | Transcription regulation | 4 | 0.32 | 1.3 | 0.119080184 |
| *ANO5* | Calcium-activated chloride channel, muscle function | 5 | 0.33 | 1.2 | 0.393877250 |
| *ZNF471* | Transcription regulation | 4 | 0.33 | 1.3 | 0.158535091 |
| *CAPN11* | Calcium-dependent cysteine protease, function in eye lens | 3 | 0.33 | 1.5 | 0.010957001 |
| *CAPRIN2* | Regulation of cell growth and proliferation | 5 | 0.33 | 1.5 | 0.083638785 |
| *ADAM32* | Cell adhesion, metalloproteinase activity | 4 | 0.33 | 0.0 | 0.172935139 |
| *ZNF605* | Transcription Regulation | 3 | 0.33 | 1.3 | 0.182883540 |
| *TSPYL2* | Regulation of cell cycle, apoptosis | 5 | 0.34 | 1.5 | 0.142211801 |
| *SAMHD1* | Antiviral defense, dNTP triphosphohydrolase | 4 | 0.34 | 1.2 | 0.364669312 |
| *RNASE9* | Ribonuclease activity | 3 | 0.34 | 0.7 | 0.499808258 |
| *FAM194A* | Unknown function | 5 | 0.34 | 1.4 | 0.210150968 |
| *DNMT1* | DNA methylation (maintenance) | 4 | 0.34 | 1.4 | 0.006395649 |
| *PVRIG* | Viral RNA sensing, innate immunity | 3 | 0.35 | 2.7 | 0.213854555 |
| *FRMPD4* | regulation of synapse development and neuronal excitability | 3 | 0.35 | 1.2 | 0.035944297 |
| *CATSPER2* | Calcium channel, sperm motility | 3 | 0.36 | 1.1 | 0.743423639 |
| *ADAM20* | Proteolysis, metalloproteinase activity | 4 | 0.36 | 0.9 | 0.644072923 |
| *LRRC70* | Unknown function | 4 | 0.36 | 2.6 | 0.063045823 |
| *ZNF550* | Transcription regulation | 4 | 0.36 | 1.3 | 0.206152967 |
| *THAP9* | DNA transposase, P-element transposition | 4 | 0.36 | 1.4 | 0.035245607 |
| *LRRIQ3* | Unknown function | 4 | 0.36 | 1.4 | 0.365890477 |
| *RLF* | Transcription regulation | 4 | 0.37 | 1.2 | 0.215214503 |
| *LRIF1* | ligand-dependent nuclear receptor-interacting factor | 3 | 0.37 | 5.7 | 0.082269966 |
| *KDM4C* | Histone demethylation, chromatin regulation | 3 | 0.37 | 1.4 | 0.013942742 |
| *ARMCX5-GPRASP 2* | Fusion gene, uncertain function | 4 | 0.37 | 1.7 | 0.000715162 |
| *ZNF74* | Transcription regulation | 3 | 0.37 | 1.2 | 0.141731853 |

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|---|---|---|---|---|---|
| *ARMCX5* | X-linked gene, potentially involved in brain development | 4 | 0.37 | 0.5 | 0.360511209 |
| *SP140L* | Nuclear body component, transcriptional regulation | 3 | 0.37 | 0.9 | 0.457961840 |
| *ZXDC* | Transcription regulation, zinc finger protein | 3 | 0.38 | 1.0 | 0.927201225 |
| *MAN2A1* | Mannosidase, glycoprotein processing | 3 | 0.38 | 1.3 | 0.116581264 |
| *SLC4A11* | Borate transport, pH regulation | 3 | 0.38 | 1.2 | 0.155896121 |
| *GPR50* | Melatonin receptor-related, orphan receptor | 3 | 0.38 | 1.5 | 0.171759410 |
| *EFCAB12* | Calcium binding protein | 4 | 0.38 | 2.7 | 0.004883126 |
| *MUM1L1* | Unknown Function | 3 | 0.38 | 1.0 | 0.969584048 |
| *MPHOSPH8* | Cell cycle regulation, M-phase phosphoprotein | 3 | 0.38 | 0.9 | 0.600241272 |
| *NRK* | Embryonic development, cell signaling | 4 | 0.38 | 1.2 | 0.149375401 |
| *ZNF470* | Transcription regulation | 3 | 0.38 | 0.8 | 0.322864119 |
| *ZNF516* | Transcription regulation | 3 | 0.38 | 1.3 | 0.008441935 |
| *ZNF227* | Transcription regulation | 4 | 0.38 | 1.0 | 0.925582087 |
| *ZNF529* | Transcription regulation | 5 | 0.39 | 1.9 | 0.164977365 |
| *YLPM1* | Unknown Function | 3 | 0.39 | 1.3 | 0.022234304 |
| *C1orf101* | Unknown Function | 5 | 0.39 | 1.2 | 0.510049215 |
| *IL4I1* | Amino acid oxidase, immune regulation | 3 | 0.39 | 1.2 | 0.182560520 |
| *ARHGAP20* | Rho GTPase regulation | 3 | 0.39 | 0.8 | 0.171141035 |
| *NME8* | Nucleoside diphosphate kinase, ciliary motility | 3 | 0.39 | 1.0 | 0.922718472 |
| *KIAA1551* | Unknown Function | 4 | 0.40 | 1.1 | 0.745746421 |
| *CXorf30* | Unknown Function | 3 | 0.40 | 1.0 | 0.961225800 |
| *CALCR* | Calcitonin receptor, calcium homeostasis | 3 | 0.40 | 1.4 | 0.269159142 |
| *PTCHD3* | Patched-related, potentially involved in signaling | 4 | 0.40 | 1.5 | 0.152369062 |
| *IQUB* | Involved in Ciliogenesis and Hedgehog Signaling regulation | 4 | 0.40 | 1.3 | 0.502572789 |
| *MANSC4* | Unknown function | 3 | 0.40 | 0.7 | 0.364497232 |
| *SPATA31D5P* | Likely non-functional pseudogene of SPATA31D family | 3 | 0.40 | 0.7 | 0.767360354 |
| *KCNU1* | Potassium channel, calcium-activated, large conductance | 3 | 0.40 | 1.1 | 0.732874612 |
| *ZNF852* | Transcription regulation | 3 | 0.40 | 1.4 | 0.116816618 |
| *ZNF774* | Transcription regulation | 3 | 0.40 | 1.3 | 0.462325439 |
| *RMDN2* | unknown function | 3 | 0.40 | 1.1 | 0.774506012 |
| *DLGAP5* | Cell cycle regulation, spindle assembly | 5 | 0.40 | 1.6 | 0.015065367 |
| *CCDC121* | Coiled-coil domain containing, unknown function | 3 | 0.41 | 0.5 | 0.616089122 |
| *ZFP28* | Transcription regulation | 4 | 0.41 | 0.8 | 0.262826219 |
| *AX747828* | Unknown, likely non-coding RNA or uncharacterized gene | 3 | 0.41 | 0.0 | 0.256451567 |
| *LRRC63* | Leucine-rich repeat containing, unknown function | 4 | 0.41 | 1.3 | 0.467494113 |
| *H1FOO* | Oocyte-specific histone H1 variant, chromatin structure | 3 | 0.41 | 0.9 | 0.748259509 |
| *FAM194B* | Unknown Function | 3 | 0.41 | 0.3 | 0.204977769 |
| *ZDBF2* | Imprinted gene, transcription regulation | 5 | 0.41 | 0.9 | 0.649565248 |
| *SLCO6A1* | Organic anion transport | 5 | 0.41 | 0.0 | 0.358771455 |
| *ZFP112* | Transcription Regulation | 4 | 0.42 | 1.5 | 0.111509962 |
| *GNPTAB* | Lysosomal enzyme targeting, GlcNAc-1-phosphotransferase | 4 | 0.42 | 0.9 | 0.489874807 |
| *NLRP5* | Maternal effect, oocyte/embryo development, inflammasome | 4 | 0.42 | 1.1 | 0.813980665 |

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|---|---|---|---|---|---|
| *ZNF77* | Transcription regulation | 3 | 0.42 | 50.0 | 0.000400878 |
| *WNK3* | Kinase, regulation of ion transport | 5 | 0.42 | 0.9 | 0.802255325 |
| *CATSPERG* | Sperm-specific calcium channel, sperm hyperactivation | 5 | 0.42 | 1.3 | 0.147615005 |
| *TNRC18* | RNA silencing, involved in miRNA pathway | 4 | 0.42 | 1.1 | 0.163685701 |
| *CATSPERD* | Sperm-specific calcium channel, sperm hyperactivation | 4 | 0.42 | 1.1 | 0.692658692 |
| *ZNF404* | Transcription regulation | 3 | 0.43 | 5.8 | 0.170352130 |
| *NOL9* | Ribosome biogenesis | 3 | 0.43 | 1.3 | 0.202469262 |
| *ZNF33A* | Transcription regulation | 3 | 0.43 | 1.7 | 0.320601709 |
| *TPTE2* | Transmembrane phosphatase with tensin homology, pseudogene | 3 | 0.43 | 0.5 | 0.272908331 |
| *CATSPERB* | Sperm-specific calcium channel, sperm hyperactivation | 4 | 0.43 | 50.0 | 0.055138043 |
| *RGAG4* | unknown function, Retrotransposon Gag Like 4 | 4 | 0.44 | 1.2 | 0.340911831 |
| *QSER1* | Glutamine and serine rich protein 1 | 4 | 0.44 | 1.1 | 0.455682370 |
| *PP2D1* | Protein phosphatase, regulation of cell signaling | 4 | 0.44 | 1.0 | 0.877608366 |
| *ZNF614* | Transcription regulation | 3 | 0.44 | 19.9 | 0.036305221 |
| *ZNF445* | Transcription regulation | 5 | 0.44 | 0.9 | 0.476669125 |
| *CEP135* | Centrosomal protein, microtubule organization | 3 | 0.44 | 1.2 | 0.245123665 |
| *WDR96* | Involved in Ribosome Biogenesis | 4 | 0.45 | 0.9 | 0.796908882 |
| *TLR8* | Toll-like receptor, innate immunity, viral RNA sensing | 3 | 0.45 | 1.3 | 0.103185022 |
| *ZMYM1* | Transcription regulation | 4 | 0.45 | 1.1 | 0.782316801 |
| *CCDC175* | Unknown function | 5 | 0.45 | 0.8 | 0.739045412 |
| *GLB1L2* | Beta-galactosidase-like protein | 3 | 0.45 | 1.0 | 0.887344045 |
| *TPTE* | Transmembrane phosphatase, implicated in various cancers | 3 | 0.45 | 0.7 | 0.338304737 |
| *CCDC8* | regulation of neuronal development | 3 | 0.45 | 0.9 | 0.643536386 |
| *FAM205A* | Unknown function | 3 | 0.45 | 1.2 | 0.400528579 |
| *AK9* | Adenylate kinase, nucleotide metabolism | 3 | 0.45 | 1.0 | 0.939956302 |
| *SGOL2* | Shugoshin, chromosome segregation | 4 | 0.46 | 1.0 | 1.000000000 |
| *ZNF268* | Transcription regulation | 4 | 0.46 | 1.6 | 0.012704098 |
| *RFPL4B* | Ret finger protein like, unknown function | 3 | 0.46 | 0.1 | 0.093711624 |
| *IRF7* | Interferon regulatory factor, antiviral response | 3 | 0.46 | 1.4 | 0.160815680 |
| *WDR87* | unknown function | 4 | 0.46 | 1.5 | 0.000180416 |
| *ZNF45* | Transcription regulation | 3 | 0.46 | 1.2 | 0.393196617 |
| *ZMYM6* | Transcription regulation | 3 | 0.47 | 1.3 | 0.062663549 |
| *ANKRD53* | unknown function | 3 | 0.47 | 0.9 | 0.737378989 |
| *ZNF483* | Transcription regulation | 3 | 0.47 | 0.8 | 0.542080213 |
| *GP6* | Platelet glycoprotein VI, collagen receptor | 3 | 0.47 | 0.6 | 0.234992853 |
| *NLRP7* | Innate immunity, inflammasome, maternal effect | 3 | 0.48 | 1.0 | 0.929407373 |
| *IL33* | Cytokine, immune response, inflammation | 3 | 0.48 | 1.4 | 0.248680561 |
| *CENPI* | Kinetochore protein, chromosome segregation | 4 | 0.48 | 2.1 | 0.079069822 |
| *CASP8* | Apoptosis (programmed cell death) | 3 | 0.48 | 0.8 | 0.562236012 |
| *PRR23A* | Proline rich protein, unknown function | 3 | 0.48 | 0.9 | 0.701545405 |
| *TET1* | DNA demethylation, epigenetic regulation | 5 | 0.48 | 1.2 | 0.249663485 |
| *POLRMT* | Mitochondrial RNA polymerase | 3 | 0.49 | 1.4 | 0.000355238 |

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|---|---|---|---|---|---|
| *CENPC1* | Centromere protein, kinetochore assembly | 4 | 0.49 | 1.1 | 0.875681737 |
| *AHRR* | Aryl hydrocarbon receptor repressor, dioxin response | 4 | 0.49 | 1.2 | 0.321927285 |
| *ZNF544* | Transcription regulation | 4 | 0.50 | 0.7 | 0.109239970 |

**Table S4.** Genes with accelerated rates on the marine lineages (≤0.50 ratio), with FDR ≤ 0.01 for the Proportional: Proportional Partitioned test and at least 3 informative branches. The genes are sorted on the estimated rate ratio (increasing). RELAX relaxation/intensification parameters K (color coded for K<1 or K>1) and p-values (color coded for p ≤ 0.00001) are reported for comparing subterranean branches with the rest of the tree. Gene process annotation automatically generated by Gemini 2.5 Pro experimental.

| Gene ID | Process | Branch Count | Relative Rate | RELAX K | RELAX p-value |
|---|---|---|---|---|---|
| *MYH16* | Muscle contraction, specifically in masticatory muscles | 6 | 5.64 | 0.33 | 0.0000000 |
| *MYH1* | Muscle contraction, skeletal muscle | 5 | 3.95 | 0.46 | 0.0000000 |
| *MYH4* | Muscle contraction, skeletal muscle | 7 | 3.89 | 0.46 | 0.0000000 |
| *NOX5* | Reactive oxygen species (ROS) production | 5 | 3.88 | 0.65 | 0.0000000 |
| *OR51B2* | Olfaction (smell) | 5 | 3.13 | 0.80 | 0.0006363 |
| *OR2T3* | Olfaction (smell) | 5 | 3.03 | 0.70 | 0.0000001 |
| *MYH13* | Muscle contraction, extraocular muscles | 6 | 2.97 | 0.52 | 0.0000000 |
| *OR52E6* | Olfaction (smell) | 6 | 2.96 | 0.92 | 0.1205441 |
| *ZNF420* | Zinc finger protein, likely transcriptional regulation | 8 | 2.91 | 0.56 | 0.0000000 |
| *OR10Q1* | Olfaction (smell) | 5 | 2.88 | 0.73 | 0.0000194 |
| *ZNF12* | Zinc finger protein, likely transcriptional regulation | 6 | 2.84 | 0.60 | 0.0000000 |
| *OR2AP1* | Olfaction (smell) | 5 | 2.67 | 1.06 | 0.4064930 |
| *OR8B3* | Olfaction (smell) | 6 | 2.63 | 0.84 | 0.0192753 |
| *KRT10* | Keratin, structural component of epidermis | 5 | 2.54 | 0.46 | 0.0000000 |
| *KRT86* | Keratin, structural component of hair and nails | 9 | 2.51 | 0.72 | 0.0000059 |
| *SSTR4* | Somatostatin receptor, involved in hormone signaling | 6 | 2.48 | 0.21 | 0.0000000 |
| *KRT2* | Keratin, structural component of epidermis | 5 | 2.45 | 0.37 | 0.0000000 |
| *HLA-C* | Immune response, Major Histocompatibility Complex (MHC) class I | 6 | 2.44 | 0.44 | 0.0000019 |
| *OR1F1* | Olfaction (smell) | 5 | 2.40 | 0.26 | 0.0000000 |
| *TCRBV6S2A1N1T* | T-Cell receptor, immune response | 5 | 2.38 | 0.99 | 0.9519340 |
| *SEC14L4* | Phospholipid binding and transport | 5 | 2.35 | 0.65 | 0.0000074 |
| *GSTM4* | Detoxification, glutathione S-transferase | 5 | 2.34 | 0.63 | 0.0002058 |
| *PRICKLE2* | Planar cell polarity, Wnt signaling pathway | 9 | 2.29 | 0.77 | 0.0000000 |
| *ADCY5* | Adenylate cyclase, cAMP signaling | 9 | 2.28 | 0.60 | 0.0000000 |
| *KRT75* | Keratin, structural component, hair follicle | 5 | 2.26 | 0.54 | 0.0000000 |
| *ZNF136* | Zinc finger protein, likely transcriptional regulation | 5 | 2.25 | 0.77 | 0.0945225 |
| *ZNF559* | Zinc finger protein, likely transcriptional regulation | 6 | 2.23 | 0.00 | 0.0030441 |
| *KRT38* | Keratin, structural component of hair | 6 | 2.21 | 0.22 | 0.0000000 |
| *MRGPRE* | G-protein coupled receptor, sensory neuron function | 5 | 2.18 | 0.67 | 0.0101004 |
| *ZNF17* | Zinc finger protein, likely transcriptional regulation | 5 | 2.16 | 0.93 | 0.3341765 |

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|---|---|---|---|---|---|
| *IGSF22* | Cell adhesion, immunoglobulin superfamily | 6 | 2.16 | 0.49 | 0.0000000 |
| *OR6C76* | Olfaction (smell) | 5 | 2.12 | 0.69 | 0.0000004 |
| *MAGEA4* | Melanoma-associated antigen, function in cancer | 5 | 2.11 | 0.87 | 0.4475649 |
| *KRT37* | Keratin, structural component of hair | 5 | 2.11 | 0.33 | 0.0000000 |
| *FBXO27* | F-box protein, part of ubiquitin ligase complex | 8 | 2.10 | 0.00 | 0.0000007 |
| *OR52N1* | Olfaction (smell) | 6 | 2.05 | 0.73 | 0.0000091 |
| *LIPJ* | Lipase, involved in lipid metabolism | 6 | 2.04 | 0.95 | 0.4892000 |
| *OR8U8* | Olfaction (smell) | 5 | 2.01 | 0.87 | 0.0774242 |
| *ZNF540* | Zinc finger protein, likely transcriptional regulation | 6 | 2.00 | 0.84 | 0.1628615 |

**Table S5.** Genes with accelerated rates on the hairless lineages (≥2 ratio), with FDR ≤ 0.01 for the Proportional: Proportional Partitioned test and at least 5 informative branches. The genes are sorted on the estimated rate ratio (decreasing). RELAX relaxation/intensification parameters K (color coded for K<1 or K>1) and p-values (color coded for p ≤ 0.00001) are reported for comparing subterranean branches with the rest of the tree. Gene process annotation automatically generated by `Gemini 2.5 Pro experimental`.

| Gene ID | Process | Branch Count | Relative Rate | RELAX K | RELAX p-value |
|---|---|---|---|---|---|
| *TRIM27* | Ubiquitination, immune response, transcriptional regulation | 5 | 0.05 | 1.76 | 0.00022543 |
| *ZNF585A* | Transcriptional regulation (likely, as it's a zinc finger protein) | 5 | 0.23 | 1.18 | 0.13449380 |
| *LOC93622* | Unknown, uncharacterized protein | 5 | 0.23 | 1.05 | 0.83473143 |
| *ANKRD30B* | Unknown function, expression in testis. Potentially transcriptional regulation. | 5 | 0.28 | 12.79 | 0.00000001 |
| *ZNF546* | Transcriptional regulation (likely, as it's a zinc finger protein) | 5 | 0.30 | 1.92 | 0.00000145 |
| *ANKRD30A* | Unknown function, expression in testis. Potentially transcriptional regulation. | 5 | 0.31 | 6.27 | 0.00000000 |
| *IGF1R* | Receptor tyrosine kinase, cell growth, proliferation, survival, differentiation | 6 | 0.32 | 1.39 | 0.00000002 |
| *MAP1S* | Microtubule-associated protein, autophagy, apoptosis | 6 | 0.33 | 1.10 | 0.05897893 |
| *ANKRD36B* | Unknown function, expression in testis. Potentially protein-protein interaction | 5 | 0.34 | 0.75 | 0.18603454 |
| *CNST* | Centrosomal protein, required for normal mitotic progression and cell division. | 5 | 0.34 | 0.65 | 0.00889621 |
| *ZNF404* | Transcriptional regulation (likely, as it's a zinc finger protein) | 7 | 0.36 | 3.60 | 0.03520551 |
| *ELF4* | Transcription factor, Ets family, hematopoiesis, inflammation | 5 | 0.37 | 1.46 | 0.00000242 |
| *ZNF470* | Transcriptional regulation (likely, as it's a zinc finger protein) | 6 | 0.37 | 0.95 | 0.65504470 |
| *ZFP82* | Transcriptional regulation (likely, as it's a zinc finger protein) | 6 | 0.37 | 1.11 | 0.21808261 |
| *POTEG* | Unknown, Potentially involved in apoptosis/ expressed in prostate, ovary, testis. | 5 | 0.38 | 0.60 | 0.23475771 |
| *IFIT5* | Interferon-induced protein, antiviral response | 6 | 0.40 | 1.27 | 0.05247376 |
| *ATP2B4* | Calcium ion pump, calcium homeostasis | 6 | 0.40 | 1.66 | 0.00000098 |
| *ZNF471* | Transcriptional regulation (likely, as it's a zinc finger | 5 | 0.42 | 1.11 | 0.37605841 |

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|---|---|---|---|---|---|
| | protein) | | | | |
| *ZNF605* | Transcriptional regulation (likely, as it's a zinc finger protein) | 6 | 0.42 | 1.30 | 0.01670276 |
| *CXXC11* | Regulation of cell proliferation and neuronal differentiation | 6 | 0.42 | 1.06 | 0.58301376 |
| *ZNF74* | Transcriptional regulation (likely, as it's a zinc finger protein) | 7 | 0.42 | 1.20 | 0.00696989 |
| *CCDC144A* | Function not fully understood. Involved in Cell motility | 5 | 0.43 | 1.50 | 0.03215042 |
| *CATSPER2* | Calcium channel, sperm motility, fertilization | 6 | 0.43 | 1.21 | 0.16000845 |
| *RBM12B* | RNA-binding protein, Potential role in RNA processing/splicing. | 8 | 0.44 | 1.48 | 0.02200487 |
| *ZNF227* | Transcriptional regulation (likely, as it's a zinc finger protein) | 7 | 0.45 | 1.14 | 0.13839699 |
| *SLC4A11* | Transporter, Bicarbonate transport, pH regulation | 5 | 0.45 | 1.06 | 0.48573376 |
| *PHF16* | PHD finger protein, function is not fully understood. May play a role in transcription regulation. | 5 | 0.45 | 1.06 | 0.55197262 |
| *ZNF133* | Transcriptional regulation (likely, as it's a zinc finger protein) | 5 | 0.46 | 1.32 | 0.00766487 |
| *LONRF3* | Regulation of ubiquitination; E3 ubiquitin ligase component | 5 | 0.46 | 1.21 | 0.03583108 |
| *CATSPERB* | Calcium channel subunit, sperm motility, fertilization | 9 | 0.46 | 1.13 | 0.57999553 |
| *ANKRD36* | Unknown function, expression in testis. Potentially protein-protein interaction | 6 | 0.46 | 1.30 | 0.00842880 |
| *ZNF197* | Transcriptional regulation (likely, as it's a zinc finger protein) | 6 | 0.47 | 1.20 | 0.01852131 |
| *POTEB* | Unknown, Potentially involved in apoptosis/ expressed in prostate, ovary, testis. | 6 | 0.47 | 1.04 | 0.84440658 |
| *ZNF230* | Transcriptional regulation (likely, as it's a zinc finger protein) | 5 | 0.48 | 1.67 | 0.02221295 |
| *C11orf95* | Unknown. Involved in various cancers as fusion with other genes. | 5 | 0.48 | 1.28 | 0.00619885 |
| *TRIM11* | E3 ubiquitin-protein ligase, involved in immune response, and regulation of protein turnover. | 7 | 0.48 | 1.23 | 0.00454031 |
| *SNX18* | Endocytosis, membrane trafficking, SH3 and PX domain-containing protein. | 7 | 0.50 | 1.26 | 0.01770941 |

**Table S6.** Genes with accelerated rates on the hairless lineages (≤0.50 ratio), with FDR ≤ 0.01 for the Proportional: Proportional Partitioned test and at least 5 informative branches. The genes are sorted on the estimated rate ratio (increasing). RELAX relaxation/intensification parameters K (color coded for K<1 or K>1) and p-values (color coded for p ≤ 0.00001) are reported for comparing subterranean branches with the rest of the tree. Gene process annotation automatically generated by `Gemini 2.5 Pro experimental.`